Giri's Tech Hub, Pune.

Core Java Machine Test

Batch: July-25

Date : 25/02/2026
Time : 2.30 to 05 Pm.

Instructions:

1. Solve any 7 questions.
2. Input should be from user.
3. Indentation and comments mandatory.
4. Without using any inbuilt functions.

Q1. Write a java program to check Number Is Armstrong Number or Not Armstrong using function recursion.

Q2. Write a java program to print this pattern.

```
A                                   A
A   B                           B   A
A   B   C               C       B   A
A   B   C   D       D   C       B   A
A   B   C               C       B   A
A   B                           B   A
A                                   A
```

Q3. Write a java program to take input array from user and perform operations in array.
Store only prime number at the same index of digit.

Input :- 5732 8659 2534 9625 7354 1325
Output :- 2573 5 253 25 357 235

Q4. Store N strings in an ArrayList. Reverse each string manually and store in another Arraylist.

Input: 3
java
code
list
Output : Reversed List: avaj edoc tsil

Description:
Traverse each string character by character in reverse order. Do not use StringBuilder reverse() or inbuilt reverse methods.

Q5. Write a java program to store words in an ArrayList. Print words that start and end with the same character.

Input: [level, cat, radar, java, madam]
Output : level , radar , madam
Description:
Traverse each word and compare first and last character manually using charAt().

Q6. Write a java program to store N integers in a Vector. Replace all even numbers with 0.

Input:
6
2 5 8 7 10 3

Output:
Modified Vector: 0 5 0 7 0 3

Description:
Traverse Vector and check modulus condition manually. Replace element using set() method.

Q7.Write a java program to store N integers in a HashMap. Find all Armstrong numbers.

Input:
6
153 10 370 89 407 25
Output:
Armstrong Numbers: 153 370 407

Description:
Store the given integers in a HashMap where the key is the index and the value is the number.
Traverse the Map values one by one.
For each number:

- **Extract digits manually using % 10 and / 10.**
- **Do not use Math.pow().**

Q8. tore N words in a HashMap. Group all anagrams together without using sort().

Input: 6
listen , silent , cat , act , dog , god
Output:
Anagram Groups:
listen silent
cat act
dog god

Description:
Store the given words in a HashMap.

Do not use:

- **Arrays.sort()**
- **Collections.sort()**
- **Any built-in anagram checking methods**

Finally, traverse the Map and print groups containing more than one word.

-----ALL THE BEST-----